The Big Pool Bangs Theory

The Big Pool Bangs Theory: Cosmology via Jammed Phase Transitions and the Hard-Sphere Kinetic Bounce

Abstract. We explore a mechanical alternative to inflationary ΛCDM in which the universe begins as a jammed ensemble of $\sim 10^{60}$ Planck-mass relics, treated as effective hard spheres (an ansatz motivated by the Compton–Schwarzschild coincidence). A loop-quantum-cosmology-type modified Friedmann equation, with critical density conjecturally identified with the hard-sphere jamming density, produces a non-singular bounce; the subsequent chaotic rebound ejects most of the ensemble. Toy N-body simulations with a turbulent kick distribution eject $88\% \pm 4\%$ of the mass, consistent with the observed dark-matter fraction for $\sigma_{\mathrm{rel}} \approx 1$ — a regime a cascade nucleation model only marginally reaches — and show clean velocity sorting of the ejecta into radial shells. The framework does *not* reproduce cosmic acceleration: coasting expansion is disfavored by Pantheon+ ($\Delta\chi^2 \approx 84$) and DES-SN5YR ($\Delta\chi^2 \approx 131$), a relativistic LTB lightcone through the ejecta decelerates ($q_0^{\mathrm{eff}} = +0.16$), a toy effective $w(z)$ drifts in the wrong direction relative to the DES+DESI fit, and a kinetic-Sunyaev–Zel'dovich gate shows that no ejection profile can supply more than $\sim 1\%$ of the observed dark energy. We therefore adopt the pool model *plus a constant tension of space* ($w = -1$) as the expansion history: the rebound sets the origin and velocity structure of the matter, the tension sets the acceleration, and the value of the tension is not derived. A predicted Cold Spot hot ring at 5ř–7ř is absent in Planck 2018 data. We organize the claims into seven hypotheses — the bounce-and-ejection core (H1), the Cold Spot as a frozen vortex (H2), a neighbouring Big Pool Bang in the Cold Spot direction (H3), a population of such neighbours (H4), our own position inside Universe 1 (H5), the location of the second Big Pool Bang (H6), and the Pool Model of the population (H7) — each with explicit falsifiers, catalogue eleven predictions (P1–P11) with the instruments required, and state the unsolved problems: causal structure of the bounce, relativistic formulation of ejection, the relativistic (hot) temperature of the ejecta, the horizon and flatness problems, and the expansion history.

1. Introduction: Three Questions

This paper is organized around three questions.

1. **Where are we?** If the universe began as a jammed ball of Planck-mass relics that rebounded and ejected most of its mass, then every observer sits somewhere in the resulting structure: near the surviving core, in the mixed mantle, or in the outer shell of fastest ejecta. The velocity sorting of the rebound (Sec. 3, Fig. 4) makes radial position a physical quantity, and only the outermost layers retain any memory of the initial geometry. Our current answer (H5, Sec. 8) is that the displaced-centre test is null to within a few Mpc, the bulk flow at $\sim$400 km s^{-1} is the only kinematic handle, and the hemispherical asymmetry is better assigned to an external neighbour than to our own offset.

2. **Where is the second Big Bang?** If our universe is one Big Pool Bang, others may exist, and a neighbour would imprint the last-scattering surface with a localized, non-Gaussian, possibly rotating spot. The Cold Spot at $(l, b) =$ (203ř, −56ř) is the candidate (H2, H3, H6, Secs. 6, 7, 9): direction known to ±3ř, comoving distance bounded below at ~ 46 Gly, mass $\sim 5 \times 10^{49}$ kg to within an order of magnitude, and a decisive curl-mode test (P9) already pre-registered. The eight extreme spots of the Planck map are the candidate list for a population of such neighbours (H4, H7, Sec. 10).

3. **How does the Big Pool Bang model work, and what does it explain?** A first-order vacuum crystallization jams into a hard-sphere state; a loop-quantum-cosmology-type bounce rebounds it; chaotic ejection sends $88\% \pm 4\%$ of the mass outward as dark matter (H1, Secs. 2–4). The rebound kinetics do *not* reproduce cosmic acceleration, so the adopted expansion history is the pool model plus a constant tension of space whose value is not derived (Sec. 5.2); the predicted Cold Spot thermal ring is absent, and the ejecta are relativistic, so "cold" dark matter is unproven (Sec. 12). We state these failures as plainly as the successes.

The rest of the paper answers the third question first, because the first two depend on it, and then returns to questions 1 and 2 with their own hypotheses, estimates, and falsifiers. Section 11 collects the eleven predictions with the instruments required.

1.1 Vacuum Crystallization and the Hard-Sphere State

Standard cosmology (ΛCDM) relies on scalar-field inflation to explain why the observable universe is flat, isotropic, and thermally uniform, at the cost of fine-tuning and of singularity theorems (Borde–Guth–Vilenkin). The **Big Pool Bangs Theory** is a mechanical alternative grounded in first-order phase-transition dynamics. Space begins in a false vacuum with energy density near the Planck limit ($\rho_{\text{vac}} \sim \rho_P$). Quantum fluctuations trigger a **vacuum crystallization** transition which, like any first-order transition, nucleates at many points: small bubbles of converted phase grow, collide, and merge, and the "bang" is their percolation into a single jammed state.

Timeline. Bubble walls propagate at $\leq c$, so percolating a $\sim 10^{20}\,\ell_P$ patch takes at least $10^{20}\,t_P = 5.4 \times 10^{-24}$ s divided by the number of nucleation sites per crossing length. The nucleation density is a free parameter, and any density sparse enough to give nontrivial merger statistics implies an assembly time many orders of magnitude above t_P; its consistency with trapped-surface avoidance (Sec. 3) is open.

Nucleation as the origin of the kick spectrum. The randomness of the merger history is inherited by the jammed core as internal turbulence, a candidate source for the stochastic kick spectrum that the rebound dynamics of Sec. 3 otherwise require as an input — provided nucleation-era velocity structure survives the collisional jammed epoch rather than thermalizing. A Poisson nucleation-and-merger simulation (vector-summed wall impulses) recovers a Maxwellian kick distribution (speed coefficient of variation 0.41–0.42), largely a central-limit consequence; the nontrivial statistic is the coherent-to-dispersive ratio σ_{rel}. Poisson nucleation is turbulence-dominated ($\sigma/\langle v_r \rangle \approx 5$–$8$) and cannot supply the coherent radial component the ejection fraction requires. A *cascade* model — a single seed bubble whose wall catalyzes further nucleation, halted by a jamming feedback at $\eta \approx 0.64$ — sweeps a conversion front outward and yields $\sigma_{\text{rel}} = 0.65$–$0.78$, robust across a $4\times$ range in trigger distance, against the ≈ 0.75–1.0 window the dark-matter fraction requires: the cascade reaches that window only marginally. Three caveats gate the chain "cascade → kick spectrum → ejection fraction": the $88\% \pm 4\%$ figure comes from Maxwellian-ansatz N-body runs and the cascade's own heavier-tailed kick field has not been fed through the N-body pipeline end-to-end;

the model retains structural choices (trigger distance, nucleation lag, wall-area weighting, the arrest $\eta = 0.64$, an unscanned impulse-collection window); and the wall-momentum efficiency is assumed universal, whereas in first-order-transition physics it is model-dependent and typically small (Espinosa–Konstandin–No–Servant).

Because quantum geometry prohibits localized infinite densities, the energy crystallizes into $N \approx 10^{60}$ irreducible **Planck-mass relics** ($m_P = \sqrt{\hbar c/G} \approx 21.8\,\mu$g), total mass $M_{\rm total} = N m_P \approx 2.2\times10^{52}$ kg (N is order-of-magnitude; the observable universe's matter mass is $\sim 1.5\times10^{53}$ kg). The relics saturate local volume up to the random close-packing limit $\eta_{\rm rcp} \approx 0.64$. With $\rho_P = m_P/\ell_P^3 = 5.2 \times 10^{96}$ kg m^{-3} the nominal jamming radius of such a ball would be

$$R_{\rm jam} = \left(\frac{3M_{\rm total}}{4\pi\rho_P\eta_{\rm rcp}}\right)^{1/3} \approx 1.2 \times 10^{-15} \text{ m},$$

but this number must not be read as the size of an isolated object: the Schwarzschild radius of 2.2×10^{52} kg is 3.3×10^{25} m, forty orders of magnitude larger, so a finite ball of this mass and density is deep inside its own horizon and would be a black hole, not a bouncing patch. The bounce of Sec. 3 is meaningful only if the jammed patch is the whole (or a super-horizon) universe described by a homogeneous FLRW-type metric, for which no exterior Schwarzschild geometry exists. We therefore use ρ_c, not $R_{\rm jam}$, as the physical parameter.

2. Planck-Mass Relics as Effective Hard Spheres

For a particle of mass m the reduced Compton wavelength $\lambda_C = \hbar/mc$ and the Schwarzschild radius $R_s = 2Gm/c^2$ coincide at $m \approx m_P$, where both equal $\sim \ell_P = 1.616 \times 10^{-35}$ m. This is the regime in which neither quantum field theory nor classical general relativity is separately valid. **We adopt, as a modeling ansatz rather than a derivation, that Planck relics possess a minimum spatial extent $\sim \ell_P$ that no interaction can compress**, and treat them as effective incompressible hard spheres. The ansatz is judged by the phenomenology it generates (Secs. 3–9); its microphysical justification is an open problem (Sec. 10).

3. The Kinetic Bounce and Chaotic Ejection (Hypothesis 1, "The Big Pool Bang")

Why hard-sphere pressure cannot bounce the collapse. Inside a homogeneous fluid the dynamics are governed by the Raychaudhuri equation

$$\frac{\ddot a}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3P}{c^2}\right).$$

Near random close packing the hard-sphere kinetic pressure diverges (free-volume equation of state, $P \propto n k_B T\,[1 - (\eta/\eta_{\rm rcp})^{1/3}]^{-1}$). In general relativity positive pressure *gravitates*: a divergent P makes $\rho + 3P/c^2$ more positive and accelerates collapse. No classical equation of state with $P > 0$ can bounce an FLRW collapse. The rebound must be sourced by quantum-geometric corrections that dominate when the ensemble reaches the jamming density $\rho_c = \eta_{\rm rcp}\,\rho_P$. The minimal object with the required property is the loop-quantum-cosmology (LQC) modified Friedmann equation,

$$H^2 = \frac{8\pi G}{3}\,\rho\left(1 - \frac{\rho}{\rho_c}\right), \qquad \rho_c = \eta_{\rm rcp}\,\rho_P,$$

which encodes a maximum attainable density covariantly: $H \to 0$ as $\rho \to \rho_c$, the contraction halts, and the effective Raychaudhuri equation flips sign. We propose the jamming limit of the relic gas as the physical origin of the ρ_c that LQC introduces axiomatically, while noting that LQC's own Immirzi-fixed value is $\rho_c \approx 0.41\,\rho_P$ versus $0.64\,\rho_P$ here — order-of-magnitude agreement, a coincidence to be explained rather than a derivation. A numerical integration (Fig. 1) confirms that the classical hard-sphere fluid overshoots the jamming density by $> 10^4$ (collapse) while the modified equation bounces at $\rho = \rho_c$; the bounce is built into the equation, not independently confirmed by it. Whether the bounce completes before a trapped surface forms around the still-homogeneous patch is an assumption (Sec. 10).

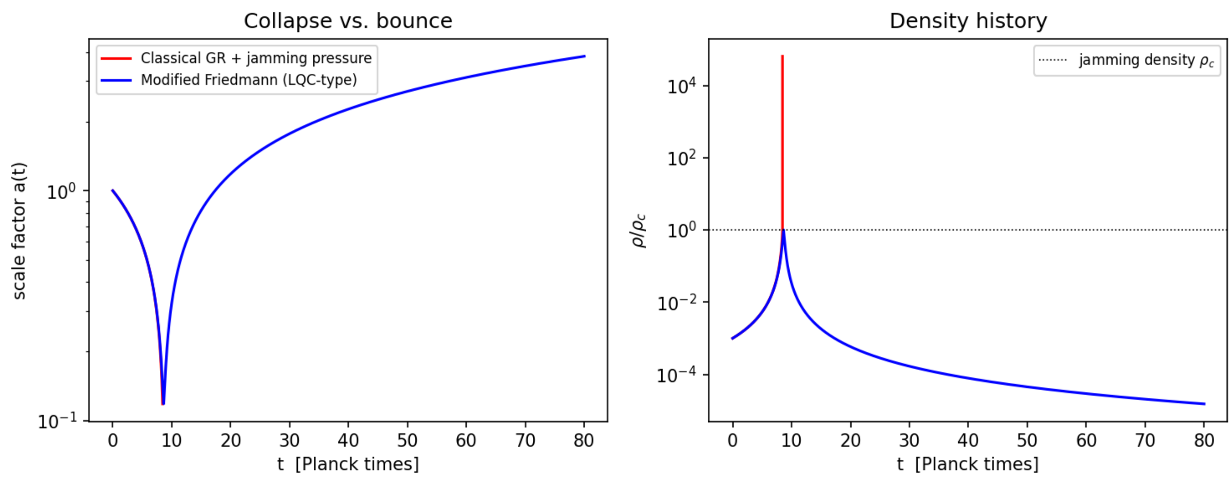

Figure 1: Fig. 1 — Toy FLRW integration. Left: scale factor. Right: density history. Classical GR with jamming pressure (red) overshoots the jamming density by $> 10^4$ (collapse); the modified Friedmann equation (blue) bounces at ρ_c.

Regime disclaimer. Everything that follows in this section is a **Newtonian analogy**: the real dynamics occur where escape velocities formally exceed c by $\sim$20 orders of magnitude and only a general-relativistic treatment (ejecta as streams on an expanding background) is meaningful. We use the Newtonian model because violent-relaxation statistics (which fraction unbinds, how ejecta sort by speed) are plausibly governed by dimensionless energy ratios that survive the translation; whether they do is the central open problem of Sec. 10.

The slingshot phase. Picture the jammed core as 10^{60} billiard balls packed to the jamming limit; when the bounce reverses the collapse, the rebound is the break. At jamming density the relic gas is collisional — mean free paths are far below a particle radius — so the slingshot phase begins only after expansion dilutes the ensemble below the contact threshold and relics decouple onto ballistic trajectories. From then on the system is a dense self-gravitating N-body ensemble in which three-body encounters dominate relaxation: two bodies drop into a tighter binary and the released binding energy ejects the third — the "core collapse and halo ejection" of globular-cluster dynamics, which sheds most of a system's mass to reach a bound remnant. Applied to the birth of the universe it sorts the mass into two components:

- **The ejected halo ($\approx 88\%$):** relics kicked above escape velocity, free-streaming through expanding space and rarely colliding again — the candidate dark-matter component. This requires no new particle, but see the temperature caveat below.

- **The trapped furnace ($\approx 12\%$):** relics that fall back and grind against each other in a turbulent core, proposed as the origin of the photon–baryon plasma. **Caveat (essential):** this identification requires an unspecified conversion channel from Planck relics to Standard-Model plasma — including baryogenesis and the photon-to-baryon ratio — while ejected relics must remain stable for 13.8 Gyr. Reconciling relic stability with furnace disintegration (e.g. a threshold collision energy the ejecta never reach) is an open requirement; absent it, the trapped fraction cannot be identified with the baryon budget.

Ejecta temperature. The ejecta leave at a substantial fraction of the (formally superluminal) escape speed, i.e. relativistically. Without a redshifting or cooling argument that brings their thermal velocities below the free-streaming limit by the epoch of structure formation, Planck-relic ejecta behave as *hot*, not cold, dark matter and would erase small-scale structure. Cosmological redshifting of free-streaming momenta ($p \propto 1/a$) is the natural candidate mechanism, but no calculation of the residual velocity dispersion at matter–radiation equality exists here; this is a first-order open problem (Sec. 10).

Ejection fraction (toy N-body). A softened $N = 1000$ equal-mass simulation of a cold uniform sphere given a radial kick $v = f\,v_{\rm esc}$ has a steep ejection threshold: $f = 0.6 \to 3\%$, $0.8 \to 41\%$, $0.9 \to 98\%$, $f \geq 1 \to 100\%$. A monolithic kick reproduces the dark-matter fraction only in the narrow window $f \approx 0.85$–0.9, and the zero-energy bounce ($f = 1$) over-ejects. A Maxwellian turbulent component of dispersion $\sigma_{\rm rel}\bar{f}v_{\rm esc}$ superimposed on a mean rebound $\bar{f}v_{\rm esc}$ broadens the transition: at $\bar{f} = 0.7$ the ejected fraction rises from 22% ($\sigma_{\rm rel} = 0.5$) through $\approx 88\%$ (1.0) to 98% (1.5). Over a 15-seed ensemble at $(\bar{f}, \sigma_{\rm rel}) = (0.7, 1.0)$ the ejected fraction is $88\% \pm 4\%$, consistent with the observed dark-matter mass fraction. This consistency holds for $\sigma_{\rm rel} \approx 1$, which the cascade model of Sec. 1 only marginally reaches (0.65–0.78); $(\bar{f}, \sigma_{\rm rel})$ remain inputs to be derived from the bounce turbulence spectrum.

Ejecta kinematic structure (15 seeds, 13{,}140 ejecta). Ejection is extended (median ejection time $\sim$1 dynamical time; 10% of ejecta leave after 11), with clean monotonic velocity sorting: median speeds fall from $0.71\,v_{\rm esc}$ for the earliest ejecta through 0.43 and 0.35 to $0.28\,v_{\rm esc}$ for the latest (Fig. 4). The fastest ejecta are outermost, so radial position maps onto ejection epoch — the input for H5 (Sec. 8). The tangential structure shows no correlated anisotropy: the per-sky ejection dipole (0.064 ± 0.024) matches the shot-noise expectation (0.053 ± 0.024), and early-vs-late shell axes align only weakly ($\langle \cos\theta \rangle = 0.37 \pm 0.49$). The isotropic cascade therefore supplies no preferred axis; if one exists it must be seeded externally (Sec. 7).

4. Consistency Checks

- **Horizon problem — open.** The jammed patch is $\sim 10^{20}$ Planck lengths across and cannot equilibrate within a few Planck times; homogeneity of the initial patch is an assumed initial condition.
- **Flatness — consistent, not explained.** A monolithic zero-energy kick ejects 100% and is incompatible with the observed split, but in the turbulent regime the simulations favor ($\bar{f} = 0.7$, $\sigma_{\rm rel} = 1.0$) the mean kinetic energy is $\langle v^2 \rangle = \bar{f}^2(1 + \sigma_{\rm rel}^2)v_{\rm esc}^2 \approx 0.98\,v_{\rm esc}^2$: total energy $K + U \approx 0$ *while* ejecting $\sim 88\%$, the positive energy of the ejecta balanced by the binding energy of the furnace. Why the rebound selects $E \approx 0$ is not derived.

Remark on the spectral tilt. One may parametrize the scalar index by a turbulent dissipation ratio ν_t

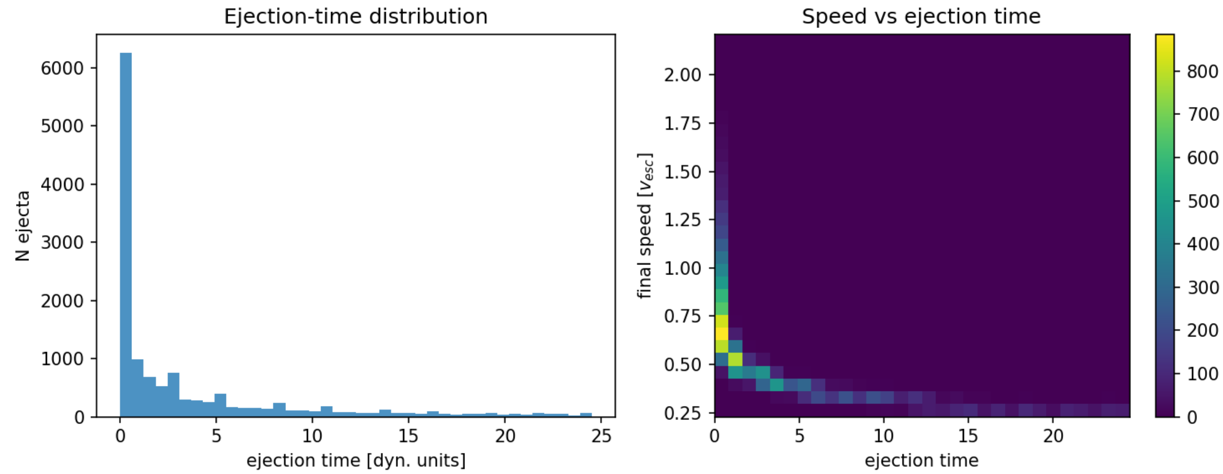

Figure 2: Fig. 4 — Ejecta kinematics over 15 seeds: ejection-time distribution (left) and speed vs ejection time (right), showing the velocity-sorted shell structure.

(eddy turnover time to expansion time) as $n_s - 1 = -\nu_t/3$; the Planck 2018 value $n_s = 0.965 \pm 0.004$ then corresponds to $\nu_t \approx 0.105$. Both the relation and the value of ν_t are fitted phenomenological inputs, not results of the framework, and showing that a Kolmogorov cascade ($E(k) \propto k^{-5/3}$) can produce a nearly scale-invariant curvature spectrum rather than a steeply red one is an open problem (Sec. 10).

5. Expansion History: Why the Pool Model Needs a Constant Tension

The rebound kinetics alone imply a coasting expansion ($a \propto t$). We confront this with data, ask whether the ejecta distribution can mimic acceleration, find that it cannot, and conclude (Sec. 5.2) that the pool model must be combined with a constant tension of space.

Supernova distance–redshift relation. Against 1371 Hubble-flow SNe Ia from Pantheon+ (diagonal errors, offset marginalized), coasting yields $\chi^2 = 674$ vs 590 for flat ΛCDM ($\Delta\chi^2 \approx 84$): disfavored, though far better than matter-only Einstein–de Sitter ($\chi^2 = 1170$; Fig. 3). On DES-SN5YR (Dovekie recalibration, 1820 SNe, full covariance): $\chi^2 = 1632$ (ΛCDM), 1763 (coasting; $\Delta\chi^2 \approx 131$), 2566 (EdS). The data require late-time acceleration beyond coasting.

Toy effective $w(z)$ from the measured shells. Integrating the measured ejecta speed distribution as cold spherical shells decelerating in their enclosed mass, and computing the equation of state an interior observer would fit, gives w_{eff} drifting from -0.29 at $z = 2$ to -0.31 today (Fig. 5): a few-percent drift near the coasting value $-1/3$, far from the observed $w_0 \approx -0.8$ in magnitude. The remap is crude — 91% of the ejecta re-bind when forced into a static enclosed-mass model, so the flow is built from the fastest 9%.

5.1 The $w(z)$ drift: evolving, but in the wrong direction

Because the apparent equation of state in this mechanism is set by the evolving ratio of velocity-dispersion energy to clustered rest-mass energy, it cannot be constant: a cosmological constant is the one behavior the mechanism cannot produce. Combined DES-SN5YR and DESI DR2 BAO analyses reject $w = -1$ at 3.2σ in a w_0w_aCDM fit ($w_0 = -0.803 \pm 0.054$, $w_a = -0.72 \pm 0.21$), with w rising

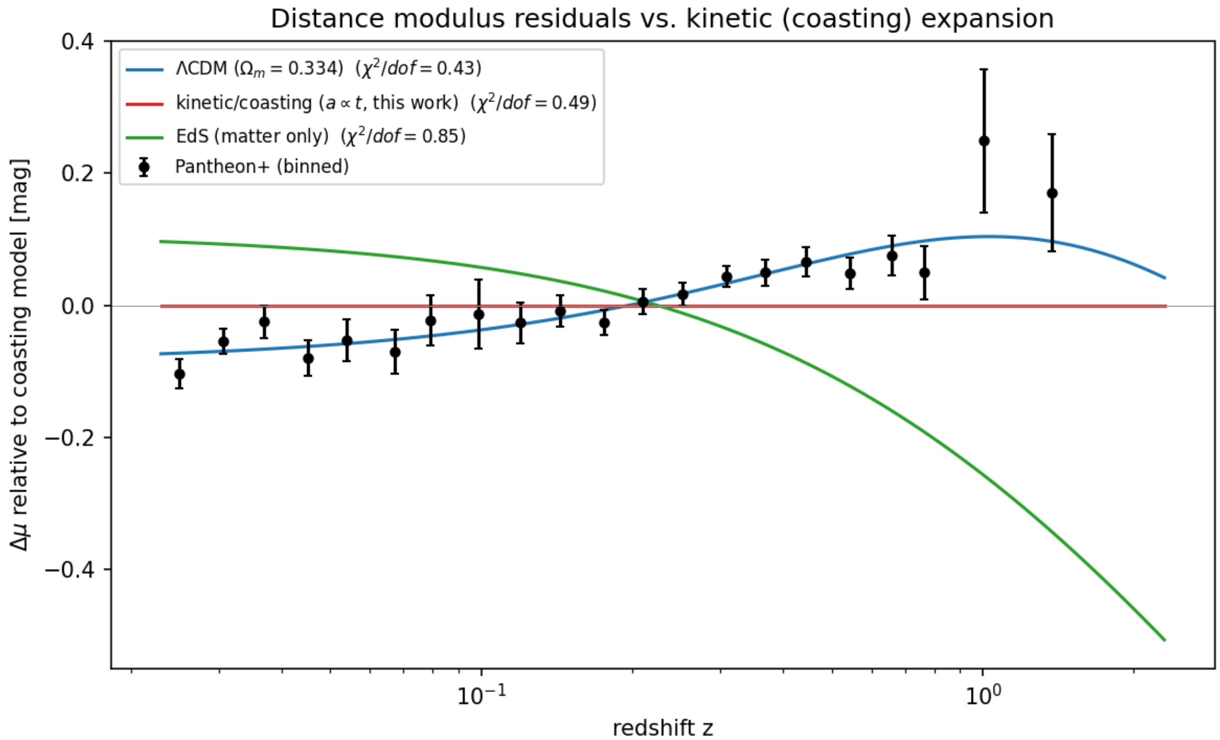

Figure 3: Fig. 3 — Pantheon+ binned distance-modulus residuals relative to the coasting model.

from ≈ -1.5 in the past toward -0.8 today (less negative with time). The toy shell calculation drifts the **opposite way** (slowly more negative with time). The framework shares with the data only the qualitative statement that dark energy evolves; its toy realization predicts the wrong drift direction, which by our own criterion counts against H1 unless a refined treatment (relativistic shells on an expanding background, furnace backreaction) reverses it.

Relativistic (LTB) recalculation. Dust shells in the Lemaître–Tolman–Bondi metric obey the same radial equation as Newtonian shells; what the toy gets wrong is the *observables*. Integrating radial null geodesics through the evolving LTB metric (300 shells seeded from the ejecta speed distribution, fallback mass virialized into the interior, $d_L = (1+z)^2 R$) gives $q_0^{\rm eff} = +0.16$ ($w_{\rm eff}(0) \approx -0.23$): **mild deceleration, no apparent acceleration**, with a distance-modulus shape $\sim$0.35 mag from both ΛCDM and coasting out to $z \sim 5$. Scope restrictions: the lightcone threads only the free-streaming 9%, the interior treatment and observer epoch are modelling choices, and shell crossing was not monitored.

Profile inversion. Since the ejection profile is an input, we searched power-law, bimodal fast-tail, and stochastically lumpy profiles for any that accelerates. Smooth families robustly decelerate ($q_0^{\rm eff} \sim +1$ to $+4$); apparently negative values in lumpy configurations are seed-ensemble noise (± 1.4, exceeding the target $|q_0| \approx 0.55$, so that family is unconstrained rather than excluded). The known way an LTB model mimics acceleration — the giant void, observer inside a Gpc-scale underdensity — is the *opposite* density ordering from outward ejection.

Blast wave and the kSZ closure. A rebound acting as a blast wave would sweep ejecta into a shell and leave a rarefied cavity, the giant-void geometry; swept-shell profiles do give smoother

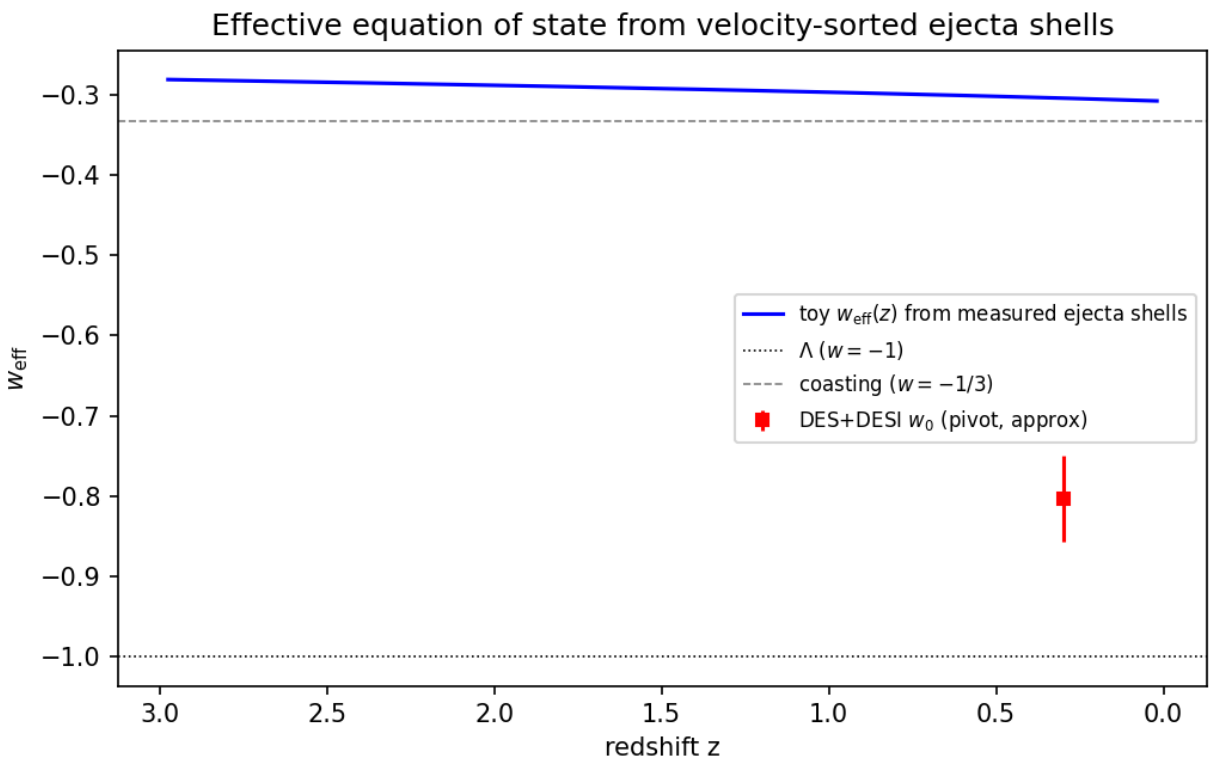

Figure 4: Fig. 5 — Toy effective equation of state from velocity-sorted ejecta shells: a few-percent drift near $w = -1/3$, in the opposite direction to the DES+DESI $w_0 w_a$ best fit, and of insufficient magnitude.

lightcones (0.03–0.08 mag residuals vs ~ 1 mag). This is not a rescue: the comparison metric cannot separate ΛCDM from coasting ($\sim$0.04 mag rms apart), a 0.08 mag residual corresponds to $\Delta\chi^2 \sim 400$ against Pantheon+, a cavity deep enough for $q_0 \approx -0.55$ is the giant-void model excluded by kinetic Sunyaev–Zel'dovich and Compton-y constraints (Zhang & Stebbins 2011), and mass swept into a distant shell is unavailable as local dark matter. Imposing the constraints first: the kSZ limit on coherent cluster flows ($v \lesssim 300$ km/s at Gpc distances) caps the mean interior contrast of any cavity at $|\bar{\delta}| \lesssim 0.05$ (500 Mpc) to $\lesssim 0.01$ (3 Gpc), i.e. $|\Delta q_0| \lesssim 0.01$ against the $\Delta q_0 \approx -1.05$ required. **No ejection profile of any morphology supplies more than $\sim$1% of the observed dark energy.** H1 must either incorporate a genuine dark-energy component or concede the acceleration sector.

A framework-native candidate, with its cost. If crystallization is incomplete today, a residual unconverted vacuum fraction would be smooth, non-clumping, and immune to the kSZ gate — qualitatively dark energy. Quantitatively this re-imports the cosmological-constant problem verbatim ($\rho_P \sim 10^{96}$ vs 6×10^{-27} kg m^{-3}: tuning to one part in 10^{123}), and ongoing conversion would imply present-epoch nucleation, domain walls (Zel'dovich–Kobzarev–Okun bound), and an unconverted component at BBN and recombination — none computed here. The boiling picture also anchors H3–H4 in Coleman–De Luccia bubble nucleation, whose collisions imprint disc-shaped CMB features with specific polarization profiles (Aguirre & Johnson) and source a stochastic gravitational-wave background (P4).

5.2 Stretch, tension, and the adopted expansion history: the pool model plus a constant tension

The failures above are failures of one idea only: that the *kinetics* of the rebound could substitute for dark energy. They do not touch the bounce, the ejection, or the dark-matter fraction. We therefore separate the two sectors explicitly. The pool model is responsible for the origin and velocity structure of the matter; the acceleration is attributed to a constant tension of space, $\rho_\Lambda c^2 = -p_\Lambda$, which we adopt without claiming to derive it.

The reason a *tension* is the only admissible form follows from a one-line argument. For a comoving separation $d = a\chi$, the second Friedmann equation gives $\ddot{d} = (\ddot{a}/a)\, d$ with $\ddot{a}/a = -\frac{4\pi G}{3}(\rho + 3p/c^2)$ (units: $G\rho \sim \mathrm{s}^{-2}$). Acceleration is therefore proportional to distance, with a coefficient set locally by density and pressure, and is repulsive only for $\rho + 3p/c^2 < 0$. If space stores an energy density $u(a)$ when stretched, thermodynamics ($p = -\partial(uV)/\partial V$) fixes its pressure: for $u \propto a^{-n}$, $w = n/3 - 1$. Matter ($n = 3$) attracts; strings or curvature ($n = 2$, $w = -1/3$) coast; a domain-wall network ($n = 1$, $w = -2/3$) repels; a constant tension ($n = 0$, $w = -1$) repels; a Hookean elastic space ($u \propto a^2$, $w = -5/3$) is phantom and excluded. The measured constant equation of state, $w = -1.03 \pm 0.03$ (CMB+BAO+SNe), selects the constant-tension case and excludes the wall network. Neighbouring Big Pool Bangs cannot contribute: a single neighbour supplies a tidal quadrupole and a bulk velocity (P5), a symmetric population supplies nothing by the shell theorem, and neither is a monopole. Today the tension term gives $\ddot{a}/a = +3.4 \times 10^{-36}$ s^{-2} against -0.7×10^{-36} s^{-2} from matter, i.e. $\ddot{d} \approx +1 \times 10^{-13}$ m s^{-2} at 1 Mpc.

Vacuum energies of neighbours do not sum. Vacuum energy acts only where it resides: the expansion rate of our region is fixed by the tension inside it, so the tensions of any number of neighbouring bangs contribute nothing here, for the same locality reason that their masses contribute no monopole (Appendix B). If all bangs crystallized into the same vacuum, the tension is one universal constant whose value the model does not predict; if into different vacua, the differences reside in the walls between them, whose network sums to $w = -2/3$ and is excluded above, while a higher-tension neighbour drives its wall outward as a bubble collision (P4).

Consequences for the rest of the paper. With the tension included, the background expansion is that of flat ΛCDM and the supernova, BAO and kSZ comparisons of this section are satisfied by construction; the ejecta shells of Sec. 3 are then a description of how the matter component acquired its Hubble flow (velocity sorting, Fig. 4) rather than an alternative expansion law. The $w(z)$ drift of Sec. 5.1 is retained as a subleading correction the kinetic sector *adds* to $w = -1$; its predicted sign is wrong and its magnitude is at the percent level, so it is not currently a usable prediction. What the framework does not supply is the value of the tension: the residual-false-vacuum estimate above reproduces the 10^{-123} tuning of standard cosmology, and we count the origin of ρ_Λ among the open problems (Sec. 12), exactly as ΛCDM does.

6. H2: The Cold Spot and the Axis of Evil (the Cold Eye of Sauron)

The Eridanus Cold Spot is the sky's outstanding non-Gaussian feature. On the Planck SMICA map it is centred at galactic $(l, b) = (203$ř$, -56$ř$)$, with a decrement of $-70\,\mu$K at 5ř smoothing and a peak of $-168\,\mu$K at 2ř smoothing, and a half-maximum radius of 2.5ř; we use these values throughout.

Cyclonic vortex reading. Internal quantum turbulence in the bounce creates macro-vortices. In the **Jovian/cyclonic model** these act as thermal dams that choke local heat transport, deform under shear into elongated features aligned along a preferred rotation axis (the "Axis of Evil"), and — without solid-surface friction to dissipate angular momentum — scale up with expansion and freeze into the CMB at decoupling. The Cold Spot is read as such a frozen cyclone.

Failed prediction: the thermal ring. A cyclone diverting thermal energy around its core was predicted to produce a concentric hot ring of +20–50 μK at 5ř–7ř. The azimuthally averaged Planck 2018 SMICA profile ($N_{\rm side} = 128$) around the Cold Spot, with uncertainties from 500 random high-latitude positions (Fig. 2), shows a cold core extending to $\sim$9ř (reaching -3.4σ at 3ř) and **no hot ring at** 5ř–7ř; a +37–52 μK excess at 13ř–15ř is $\sim 2.5\sigma$ before a trials factor of ~ 10 radial bins and is unremarkable. The thermal-ring prediction failed (P1). The polarization part survives: a frozen vortex boundary should carry a radial E-mode ring of amplitude 1–3 μK at $\sim$ 5ř–15ř (bracketing the half-max radius and the outer edge of the cold core). Planck's Q_r profile is consistent with zero in every 2ř ring to 20ř with per-ring errors of 0.6–2.7 μK, so the ring is neither detected nor excluded; the test requires CMB-S4 sensitivity.

Preferred axis. Our simulations find ejection isotropic within shot noise (Sec. 3), so the isotropic cascade alone provides no preferred axis; in this framework an axis can only be supplied by an external seed, the neighbouring bang of Sec. 7. The literature's cluster of tentative, individually sub-decisive ($\lesssim 3\sigma$) alignments — a supernova acceleration dipole roughly along our CMB motion (Colin et al. 2019; Sah & Rameez 2024), the quadrupole–octupole alignment with the dipole direction, coincident bulk-flow and quasar-polarization axes (Hutsemékers et al.) — is motivation for H2/H3, not support.

Galaxy handedness. Claimed spin-handedness dipoles (Longo 2011 at $(52$ř$, +68.5$ř$)$; Shamir 2012–2022 at $(192$ř$, +38$ř$)$) lie 70ř from each other and 50ř–74ř from the axes above; we treat them as systematics-limited (Iye et al. 2021). The JWST/JADES field at $(224$ř$, -54$ř$)$, 8.6ř from the Cold Spot, shows a claimed 2 : 1 handedness asymmetry in 263 high-redshift galaxies (Shamir 2025); we cite this only as a coincidence awaiting the control-field test (Appendix A).

Falsifier. Absence of the E-mode ring at CMB-S4 sensitivity eliminates H2. A localized curl (P9) detection at the Cold Spot would support the vortex reading over the neighbour reading of the same

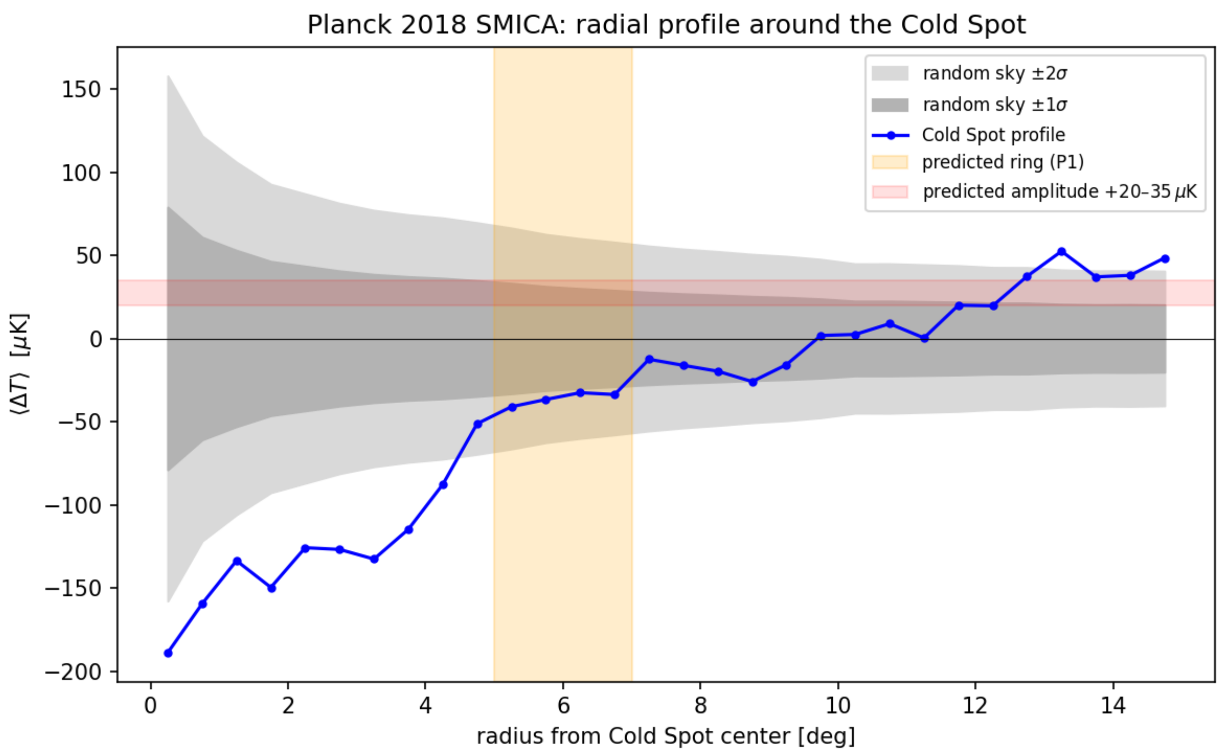

Figure 5: Fig. 2 — Planck 2018 SMICA radial temperature profile around the Cold Spot, vs. 500 random reference positions.

direction (Sec. 7); a calibrated curl null retires it.

7. H3 and H4: A Neighbouring Big Pool Bang in the Cold Spot Direction (Big Pool Bang Multiverse 2), and a Population of Neighbours (the Dark Comic Background)

Because vacuum crystallization is stochastic across an unbounded background, our bang need not be unique. A neighbouring universe nucleating beyond our domain boundary imprints its boundary on our last-scattering surface, which lies at a comoving distance of ~ 46 Gly; a neighbour visible as a CMB feature is therefore at least that far in comoving terms, and any neighbour closer than last scattering would appear in large-scale-structure tracers, which the Cold Spot sightline does not show beyond the too-shallow Eridanus supervoid.

Cross-boundary signatures. If black holes from Universe 2 cross the boundary into our sector, their properties differ from local objects:

Physical feature	Universe 1 (local)	Universe 2 (cross-boundary infall)
Mass profile	Astrophysical spectrum plus any Planck-relic clusters	Standard stellar/supermassive spectrum, with sub-solar anomaly gaps (0.1–1.0 $M_\odot$)
Sky distribution	Isotropic	Clustered toward the Cold Spot direction
Peculiar velocity	Bound to local orbital speeds (~ 200 km/s)	Directed component from the Cold Spot, bounded above by the kSZ limits ($\lesssim 300$ km/s coherent)
GW waveform	Standard inspiral chirp	Gravitational-wave memory: step-function strain $\Delta h \sim 10^{-22}$–10^{-21}

Detecting the memory strain requires isolating a non-oscillating step from noise: ground interferometers lack the sub-10 Hz response, so this is an Einstein Telescope / LISA measurement (P4).

The spin constraint. A neighbour is the only agent that can torque our universe (internal angular momentum cancels), so the measured non-rotation constrains H3. An equal-mass neighbour — taking 1.5×10^{53} kg, the observable-universe matter mass, as the upper bound for Universe 2 — at $D \sim 46$ Gly exerts a tidal field $T \sim GM/D^3$ which, acting on our quadrupole asymmetry ($q \sim 10^{-5}$) over the causally available few Gyr, induces vorticity above the Planck Bianchi bound $\omega/H_0 \lesssim 10^{-9}$ (Saadeh et al. 2016) unless the boundary transmits $\lesssim 0.1\%$ of the shear. **The Cold Spot's own amplitude fixes the mass.** If the decrement is the neighbour's Sachs–Wolfe imprint, $\Delta T/T \approx 2.6 \times 10^{-5}$ gives $M_2 \approx 3(\Delta T/T)\, c^2 D/G \approx 5 \times 10^{49}$ kg at $D = 46$ Gly — a *dwarf* neighbour, $\sim 2 \times 10^{-3}$ of M_{total}. Fed back into the torque estimate, $\omega/H_0 \sim 10^{-11}$, two orders under the bound. The mass the amplitude demands is automatically small enough to satisfy non-rotation, subject to $\mathcal{O}(1)$ Sachs–Wolfe factors and the uncomputed boundary transmission (which pushes M_2 up). The angular-size check passes: 5ř at 46 Gly subtends ~ 4 Gly.

The bubble spectrometer. The two observables of any anomalous spot invert into the neighbour's parameters: Sachs–Wolfe gives $M/D \propto \Delta T$, and an imprint scale set by the neighbour's size ($R \propto M^{1/3}$ at fixed formation density) gives $\theta \propto M^{1/3}/D$, hence

$$M \propto (\Delta T/\theta)^{3/2}, \qquad D \propto \Delta T^{1/2}\,\theta^{-3/2}.$$

These are scaling relations: absolute values require the neighbour's formation density, so the inversion supplies ranks, and the Sachs–Wolfe estimate supplies the absolute mass at an assumed distance. They apply only to the non-Gaussian outlier population — the acoustic-peak spots are ΛCDM's best-confirmed physics. Neighbours on a common distance shell populate a locus $\Delta T \propto \theta^3$, whereas Gaussian peaks obey BBKS statistics with amplitude and size nearly independent. The inferred neighbour mass function is the nucleation size spectrum of the vacuum transition — the same distribution that drove the cascade of Sec. 1 — so the internal (ejecta) and external (spot) samples of one process must agree, a cross-consistency requirement linking H1 and H4.

Southern one-sidedness. All eight most-extreme large-scale spots on the SMICA map ($|b| > 25$ř, 2ř smoothing; Appendix A) lie in the southern galactic hemisphere; in 200 Gaussian realizations through the same pipeline none placed its full top-8 in one hemisphere (mock mean 5.2 of 8). The honest significance is that of the published hemispherical power asymmetry ($p \sim 1$–10%, estimator-dependent), since our mocks are simplified and the frame post hoc. In this framework the asymmetry is not a separate anomaly: a single massive neighbour to the galactic south — the object invoked for the Cold Spot — modulates large-scale power on one side of the sky by tidal and Sachs–Wolfe action, so the asymmetry, the Cold Spot, and the preferred axis of Sec. 6 become one prediction with one direction. An isotropic swarm of neighbours (H4) predicts no such one-sidedness.

H4. Since nucleation is stochastic, the neighbour should not be unique: a population of Big Pool Bangs at varying distances, each imprinting a distinct anomaly (P7). Planck 2018 isotropy analyses find the sky consistent with Gaussian ΛCDM apart from 2–3σ large-scale anomalies and the single Cold Spot, so additional neighbours must imprint below Planck's per-spot threshold (more distant or weaker boundaries). H4 stands or falls on population statistics (Sec. 9, H7), not on any single spot.

Falsifiers. H3 is eliminated by the absence of anisotropic sub-solar mergers and memory strain at ET/LISA sensitivity, or by a foreground structure shown to account for the full decrement; H4 by a null in sub-threshold spot statistics at LiteBIRD/CMB-S4 depth.

8. H5: Where Are We Relative to Our Own Big Pool Bang 1?

Unlike standard cosmology, this framework has a centre (the seed bubble / furnace) and a radial structure — the velocity-sorted shells of Sec. 3 — so "where are we inside Universe 1: near the centre, in the mantle, or in the border shell?" is a meaningful question.

What the model expects. Observers are made of furnace material and so sit near the centre by construction; the fastest ejecta are outermost, so radial position maps onto ejection epoch (Fig. 4), and the border shell — which is what the CMB sky *is*, seen from near the centre — should carry the fossil arrangement of the earliest ejecta. Only those escaped in a single pass and preserve geometric correlations; everything slower underwent violent relaxation, which conserves spin magnitude but randomizes spin geometry. The model therefore predicts a **coherence gradient**: fossil geometry intact at the border, degrading inward, absent in the well-mixed interior. Intermediate-shell tracers (quasar-polarization alignments at $z \sim 1$–2; galaxy handedness axes) lie 24ř–73ř from the Cold Spot direction, consistent with random axes, and the local same-handedness correlation over 77{,}840 SDSS galaxies is zero to a few parts in 10^3 — as mixing requires.

Displacement from the centre. The machinery is standard (off-centre observer constraints from the CMB dipole: Alnes & Amarzguioui 2006; tilted-universe program: King & Ellis 1973; Asvesta et al. 2022); what is specific here is the attribution of a shared vector to displacement from a physical origin. Attributing the full CMB dipole (370 km/s) to off-centring gives $\Delta r \approx 5$ Mpc; the residual unexplained by local gravity gives $\Delta r \lesssim 1$–2 Mpc. A displaced centre is invisible in a linear flow but enters at $\mathcal{O}(Hc^2/r)$, ~ 90 km/s at 10 Mpc for a 5 Mpc offset. Fitting the 2{,}112 Cosmicflows-4 groups at 1.5–40 Mpc with $v(R) = hR + kR^2$ diverging from a free centre gives $|\vec{c}| = 6.1$ Mpc toward (118ř, −24ř) at $\Delta\chi^2 = 6595$, and independent shells at 40–80 and 80–150 Mpc return centres within 10ř–15ř of it. Two controls dismantle this. *Mock mask:* ten isotropic mocks on CF4's exact sky positions and distance errors manufacture false centres (0.4–2.5 Mpc, $\Delta\chi^2 \approx 9$–61 per shell) that in the outer shells point into the *same longitude band* ($l \approx$ 93ř–152ř) as the data. *Structure subtraction:* removing the 2M++ reconstruction-predicted peculiar velocities (Carrick et al. 2015) from every group drops the inner shell to $\Delta\chi^2 = 182$ ($|\vec{c}| = 1.7$ Mpc, within 8ř of the CMB dipole, plausibly a residual of the reconstruction's external-dipole calibration), the 40–80 Mpc shell to the mock noise floor (23 vs 24), and the 80–150 Mpc shell to $\sim 1.5\times$ mock. **Current answer: no displaced expansion centre is detected once mapped structure is subtracted**; any offset is $\lesssim$ a few Mpc, $\sim 10^{-4}$ of the distance to last scattering.

The only kinematic handle: the bulk flow. The bias-corrected Cosmicflows-4 bulk flow (Watkins et al. 2023) is 419 ± 36 km/s at $200\,h^{-1}$ Mpc toward (298ř ± 5ř, −8ř ± 4ř), rising with depth, in nominal $\sim 4.8\sigma$ tension with ΛCDM (Whitford et al. 2023 argue the errors are understated); Planck's kSZ analysis limits coherent flows to $\lesssim 254$ km/s at 95% on larger scales, so amplitudes $\gtrsim 600$ km/s are disfavored (P5). The flow axis and the CMB dipole (264ř, +48ř) lie 40ř–50ř apart, which a single offset vector does not predict; local-attractor infall explains the flow without any offset.

Hemispherical asymmetry as an off-centre signature? An off-centre observer in a velocity-sorted universe would see a power asymmetry along the offset axis. The observed southern one-sidedness (Sec. 7) is 90ř–115ř from every flow axis, so it is not the offset signature; we attribute it instead to the neighbour (H6). The offset vector (H5) and the Cold Spot axis (H6) are two distinct directions, and cross-boundary infall from the Cold Spot cannot be the source of the bulk flow.

Quantity	Estimate	Method	Status
Radial zone	Central furnace region	Furnace origin of baryons; coherence gradient	Model expectation, untested
Offset from centre	$\lesssim$ 2–5 Mpc; no detection	CMB dipole cap; CF4 free-centre fit with mock-mask and 2M++ controls	Null
Offset direction	Undetermined (CMB dipole (264ř, +48ř) vs flow (298ř, −8ř))	Dipole/flow attribution	Misaligned by 40ř–50ř
Bulk flow (P5)	419 ± 36 km/s at $200\,h^{-1}$ Mpc; $\gtrsim 600$ disfavored	Cosmicflows-4, kSZ	Persistence with depth untested

Quantity	Estimate	Method	Status
Coherence gradient	Interior mixed; border fossil untested	SDSS handedness null; P9 curl at the border	Interior null; border open

What would decide it. (i) The deep flow persisting, not converging, at bias-corrected depths $\gtrsim 200\,h^{-1}$ Mpc; (ii) a dipole-locked redshift *quadrupole*, whose sign fixes which end of the axis holds the centre; (iii) a mask-marginalized, structure-subtracted centre fit on number-count dipoles, pairwise velocities, and field-level inference with a free centre parameter, all requiring mocks with correlated velocity fields; (iv) the P9 curl test at the border. A persistent flow with a coherent quadrupole would place us a few Mpc off-centre; continued nulls leave the centre a hypothesis without an observable signature at distance-catalogue depth.

9. H6: Where Is the Second Big Pool Bang?

H6 gathers the neighbour claims of Sec. 7 into a location. **Direction:** the Cold Spot, $(l, b) =$ (203ř, −56ř), uncertain by $\sim$2–3ř (the half-max radius), in the hemisphere of the eight-spot one-sidedness, which we read as the neighbour's tidal and Sachs–Wolfe modulation. **Distance:** at least the comoving distance to last scattering, $\sim$46 Gly (14 Gpc), since the imprint sits on that surface and no adequate foreground counterpart exists; the spectrometer relation gives only a rank, so the upper end is unconstrained. **Mass:** $M_2 \approx 5 \times 10^{49}\,(D/46\,\text{Gly})$ kg from the Sachs–Wolfe inversion, uncertain by an order of magnitude, i.e. $\sim 10^{-3}$ of our 2.2×10^{52} kg; the equal-mass value 1.5×10^{53} kg is an upper bound excluded by the spin constraint without $\lesssim 0.1\%$ boundary transmission. **Kinematics:** the bulk flow points 90ř–115ř away from the Cold Spot, so the neighbour contributes no detected flow; any coherent component along its axis is bounded by kSZ at $\lesssim 300$ km/s. **Our position relative to it:** $\gtrsim 46$ Gly comoving from its boundary along the axis toward (203ř, −56ř); the internal offset of H5 ($\lesssim$ few Mpc) is $\sim 10^{-4}$ of that and negligible.

Quantity	Estimate	Method	Status
Direction	(203ř, −56ř) ± 3ř	Cold Spot centroid; southern one-sidedness	Fixed by data; attribution hypothetical
Distance	$\gtrsim 46$ Gly comoving	Last-scattering surface; no foreground counterpart	Lower bound only
Mass M_2	$\sim 5 \times 10^{49}$ kg (±1 dex)	Sachs–Wolfe inversion of $-70\,\mu$K	Consistent with spin bound
Induced vorticity	$\omega/H_0 \sim 10^{-11}$	Tidal torque on $q \sim 10^{-5}$	Under Saadeh et al. bound
Directed flow	$\lesssim 300$ km/s along axis; none detected	Cosmicflows-4, kSZ	Null
Curl excess at Spot (P9)	Predicted zero (neighbour) vs excess (vortex)	Lensing curl / B-mode stack	Cheap estimator null; calibrated test pending

Quantity	Estimate	Method	Status
Polarization ring	1–3 μK at 5ř–15ř	Q_r profile	Unconstrained by Planck

What H6 predicts and what falsifies it. The neighbour reading predicts a *non-rotating* imprint: zero curl at the Cold Spot at calibrated sensitivity, a disc-shaped edge profile with the Aguirre–Johnson polarization signature, anisotropic sub-solar mergers and memory strain from that direction, and no proper motion (any intrinsic transverse motion is nanoarcseconds per century). It is falsified by a curl detection at the Spot (which transfers the direction to H2), by a foreground structure accounting for the full decrement, by the absence of the bubble-collision polarization profile at CMB-S4 sensitivity, or by the one-sidedness proving an estimator or foreground artifact in end-to-end simulations. P9 is the decisive near-term test: it separates the two readings of the same direction.

10. H7: The Pool Model Hypotheses

The Pool Model gathers the population claims of H3–H6 into explicit statements.

(a) Universe 1 is one ball on the table. Vacuum crystallization nucleated more than once; our bang is one of several Big Pool Bangs separated by domain boundaries, each with its own furnace and ejected halo.

(b) Each neighbour imprints one spot. A neighbour's boundary imprints a spot on our last-scattering surface with $M \propto (\Delta T/\theta)^{3/2}$ and $D \propto \Delta T^{1/2}\theta^{-3/2}$. The eight most extreme large-scale spots on the SMICA map are therefore the candidate neighbour list. The inverted ranks below are model-dependent (common formation density, pure Sachs–Wolfe imprint), in arbitrary units normalized within the sample:

Rank	(l, b)	Sign	ΔT (μK)	$\theta_{1/2}$	Relative M	Relative D
1	(157ř, −71ř)	hot	+170	2.5ř	1.00	1.00
2	(80ř, −33ř)	cold	−170	3.5ř	0.60	0.60
3	(203ř, −56ř) (Cold Spot)	cold	−168	2.5ř	0.99	0.99
4	(155ř, −29ř)	cold	−163	5.5ř	0.29	0.30
5	(305ř, −29ř)	hot	+153	2.5ř	0.86	0.95
6	(211ř, −35ř)	cold	−153	2.5ř	0.86	0.95
7	(170ř, −47ř)	hot	+162	4.5ř	0.39	0.40
8	(184ř, −54ř)	hot	+155	4.5ř	0.36	0.40

Hot spots require a boundary imprint of the opposite sign (an overdense or approaching wall); the model does not yet specify which neighbours imprint hot. Two of the hot extremes lie 11ř–12ř from the Cold Spot and may be pieces of one system.

(c) Population predictions. Neighbours cluster on the sky: they should share the southern one-sidedness direction (currently 8/8 south). Neighbours on a common distance shell should follow the locus $\Delta T \propto \theta^3$ rather than Gaussian-peak (BBKS) statistics: the current locus test returns a slope $d\ln|\Delta T|/d\ln\theta = 0.00$ ($r = -0.01$), at the 27th percentile of the Gaussian null ($+0.07 \pm 0.10$) — null, but powerless as implemented, because the 2ř smoothing quantizes radii to 2.5ř–5.5ř and the top-8 selection compresses the amplitude spread to $\sim$10%, so a flat slope is expected under both hypotheses. Any genuine neighbour is non-rotating and should show zero curl; conversely any spot that is a relic vortex of our own bang (H2) should show P9 curl excess — the population divides on this test. The inferred neighbour mass function must match the nucleation spectrum inferred from our own ejecta kinematics.

Quantity	Estimate	Method	Status
Candidate neighbours	≤ 8 (top-8 extremes; 6 if the Cold Spot system is merged)	SMICA extrema, 2ř smoothing	Candidate list
One-sidedness	8/8 south; 0/200 mocks	Hemisphere count	Literature-level significance ($p \sim$ 1–10%)
Locus slope	0.00 vs +3 (locus) or $\sim$0 (Gaussian)	Log–log fit, 60 Gaussian nulls	Null, powerless; injection-recovery pending
CMB-independent mass counterpart	$+2.7\sigma$ sign-weighted (WISE$\times$SuperCOSMOS); null in Quaia	Galaxy-density stacks	1–2σ-class hint
Curl at 8 spots	82nd percentile (uncalibrated)	Temperature-only quadratic estimator	Null at low sensitivity

(d) What falsifies the Pool Model as a whole. All three of: the extreme spots consistent with Gaussian BBKS peaks under a validated injection-recovery locus test (a $\Delta T \propto \theta^3$ population injected and demonstrably recoverable, yet absent in data); zero curl at calibrated sensitivity at all eight spots *and* no bubble-collision polarization profile at any of them at CMB-S4 depth; and no CMB-independent mass counterpart at any spot in a pre-registered galaxy-density re-run. Under those three nulls the spot population is Gaussian, the Cold Spot is a single outlier of unknown origin, and H3–H4 and H6–H7 fall together while H1, H2, and H5 stand.

(e) Quantitative falsifiers of a surrounding population. A neighbour that imprints a compact spot cannot imprint *only* a spot. Modelling each spot as the Sachs–Wolfe signature of a point mass at comoving distance $D = (1+\epsilon)\, d_{\rm LS}$ just beyond the last-scattering sphere, the profile is $\Delta T(\theta) \propto [\epsilon^2 + 2(1+\epsilon)(1-\cos\theta)]^{-1/2}$, so the half-maximum angle fixes $\epsilon = \theta_{1/2}/\sqrt{3}$: the eight extremes require $\epsilon = 0.025$–0.055, i.e. sources 1–2.5 Gly (comoving) beyond last scattering, and by the Sachs–Wolfe amplitude a mass $M \approx 3 \times 10^{48}$ kg for the Cold Spot (within the ± 1 dex of Sec. 9). Three consequences follow with no free parameters.

1. *A low-multipole tail (P10).* The $1/r$ potential of each source extends over the whole sky at $\sim$2% of its peak: 3–6 μK at 90ř and 2–4 μK at the antipode. Summing the eight tails predicts a fixed $\ell = 1$–3 template with rms 4.1 μK at $\ell = 2$ and 4.5 μK at $\ell = 3$, i.e. $\sim$20% of the observed quadrupole power ($C_2 \approx 210\,\mu\mathrm{K}^2$) and $\sim$8% of the octopole. This is allowed, and it is a prediction: the template must correlate positively with the observed $\ell = 2, 3$ maps. A first unmasked SMICA check gives $r = +0.15$ at $\ell = 2$ and $r = -0.27$ at $\ell = 3$, and the template's quadrupole and octopole axes lie 78ř and 52ř from the observed axes — no support, not yet a rejection, since the unmasked low-ℓ map is foreground-contaminated. A masked, component-separated low-ℓ comparison (Planck Commander $\ell \leq 3$) is the pre-registered test: a correlation coefficient consistent with zero or negative at both multipoles, with the template amplitude fixed by the spots, falsifies the population reading of the extremes.

2. *A tidal bulk-flow direction (P11).* The same masses pull on the local volume; the predicted flow direction is the mass-weighted vector sum of the spot directions with weights $\Delta T\,\epsilon/(1+\epsilon)^2$. Taking cold spots as attractive (positive mass, the only sign the Sachs–Wolfe model supplies), the prediction is $(l, b) = (110ř, +24ř)$, which is 161ř from the CosmicFlows-4 bulk flow at $(297ř, -6ř)$: anti-aligned. Reversing the sign convention (hot attractive) gives $(290ř, -24ř)$, 19ř from CF4. The model therefore either fails this test or requires hot extremes to be the attractive ones, which is the unspecified sign mechanism noted under the table above. We pre-register the test as: once the sign mechanism is fixed *by the model*, the predicted direction must fall within 30ř of the measured bulk flow; the sign may not be chosen from the data.

3. *Expansion-shear anisotropy.* The Cold Spot source alone exerts a tidal field $GM/D^3 \approx 2\times10^{-42}$ s^{-2}, a fractional expansion anisotropy of $\sim 4\times10^{-7}$ of H_0^2: undetectable in supernova anisotropy searches (percent level) and below the low-ℓ Grishchuk–Zel'dovich limit by construction, so this is a consistency check the picture passes rather than a discriminant.

Finally, a genuinely *surrounding* population should populate both hemispheres. The present 8/8 southern count says the candidate population is one-sided; if the extremes are neighbours they are clustered on one side of us, not around us. The pre-registered extension is the hemispheric balance of the next twenty extremes ranked 9–28: a surrounding population predicts isotropy, a one-sided population predicts a persistent southern excess, and a Gaussian sky predicts isotropy with the top-8 count a fluctuation.

11. Falsifiable Predictions and Required Instrumentation

Hypothesis hierarchy. H1 — "The Big Pool Bang" (core, must hold): bounce plus chaotic ejection producing the dark-matter component (P6, P3); the framework does not supply dark energy (Sec. 5). **H2 — "The Cold Eye of Sauron" (bonus):** the Cold Spot as a frozen vortex and the Axis-of-Evil alignment (P1, P2, P9). **H3 — "Big Pool Bang Multiverse 2" (bonus):** a neighbouring universe in the Cold Spot direction (P3, P4). **H4 — "The Dark Comic Background" (generalizes H3):** a population of neighbours (P7). **H5:** our position inside Universe 1 (P5, P8). **H6:** the location of the second Big Pool Bang (P4, P9). **H7 — "The Pool Model":** the population statements of Sec. 10 (P7, P9). H2–H7 are separable: any can fail without falsifying the core mechanism, and each can hold without the others.

#	Prediction	Observable and required threshold	Instrument	Status
P1	Cold Spot ring. Thermal ring of $+20$–$50\,\mu$K at 5ř–7ř: **failed** in Planck 2018 (Sec. 6). Surviving prediction: radial E-mode ring of 1–3 μK at 5ř–15ř	Polarization sensitivity $\lesssim 1\,\mu\mathrm{K}\cdot$arcmin at $\ell \sim 20$–100 over a $\gtrsim$ 30ř patch	CMB-S4	Thermal: failed. Polarization: open
P2	Cyclonic B-mode pattern aligned with the Axis of Evil	Degree-scale B-modes, r-equivalent $\sim10^{-3}$	CMB-S4, LiteBIRD	Open

#	Prediction	Observable and required threshold	Instrument	Status
P3	Sub-solar mergers (0.1–1.0 $M_\odot$) clustered toward the Cold Spot	Strain $\sim 10^{-24}/\sqrt{\mathrm{Hz}}$ at 10–10^3 Hz; localization $\lesssim 10\,\mathrm{deg}^2$	LIGO/Virgo/KAGRA O5, Einstein Telescope, Cosmic Explorer	Open
P4	Gravitational-wave memory (step strain $\Delta h \sim 10^{-22}$–10^{-21}) from cross-boundary infall, plus a first-order-transition stochastic background (broken power law peaked at the transition scale). Ground detectors are blind below 10 Hz (seismic wall), so single-event memory needs sub-Hz response	$\Delta h \lesssim 10^{-22}$ below 10 Hz; memory via event stacking meanwhile	Einstein Telescope (underground, to $\sim$3 Hz), LISA (mHz), pulsar timing arrays (nHz)	Open
P5	Coherent bulk flow of $\sim$400 km/s persisting (not converging) at bias-corrected depths $\gtrsim 200\,h^{-1}$ Mpc toward $l \approx 298$ř; $\gtrsim 600$ km/s disfavored by the Planck kSZ limit	Peculiar-velocity survey to $z \sim 1$ with $\lesssim 300$ km/s per cluster	Rubin/LSST + Euclid + kSZ	Current data lean toward persistence; errors disputed
P6	No particle dark-matter candidate (DM = Planck relics)	Continued nulls approaching the neutrino floor	LZ, XENONnT	Ongoing

#	Prediction	Observable and required threshold	Instrument	Status
P7	A sub-threshold population of Cold-Spot-class anomaly spots (Planck already limits any above-threshold population)	Sub-threshold spot statistics; per-candidate polarization at $\lesssim 1\,\mu\mathrm{K}\cdot\mathrm{arcmin}$	LiteBIRD, CMB-S4	Constrained; open below threshold
P8	Spin surcharge: filaments and clusters carry primordial angular momentum atop tidal spin, excess largest at high z; randomly oriented (zero net), consistent with $\omega/H_0 \lesssim 10^{-9}$. Wang et al. (2021) filament spin is the $z \approx 0$ anchor	Filament-spin stacking at $z \sim 1$ vs $z \sim 0$ against ΛCDM torque mocks	DESI, Euclid, 4MOST	Open
P9	Localized curl excess (lensing curl and B-mode variance) at the extreme-spot positions if they are relic vortices, zero mean globally; a calibrated stacked null at the Cold Spot retires the vortex reading in favor of the neighbour reading (Appendix A)	Planck PR4 curl reconstruction at full depth; degree-scale B-modes at $\lesssim 1\,\mu\mathrm{K}\cdot\mathrm{arcmin}$	Planck PR4 (archival), CMB-S4, LiteBIRD	Cheap estimator null; calibrated test pending

#	Prediction	Observable and required threshold	Instrument	Status
P10	Low-multipole tail of the spot population: a fixed $\ell = 1$–3 template (rms $4\,\mu$K at $\ell = 2$, $4.5\,\mu$K at $\ell = 3$) summed from the eight point-mass profiles must correlate positively with the observed $\ell = 2, 3$ maps (Sec. 10e)	Masked Commander low-ℓ maps; correlation coefficient with the template at fixed amplitude	Planck 2018/PR4 (archival)	Unmasked first look: $r = +0.15$ ($\ell = 2$), -0.27 ($\ell = 3$); no support
P11	Tidal bulk-flow direction: the mass-weighted vector sum of the spot directions must fall within 30ř of the measured bulk flow once the model fixes the attractive sign (Sec. 10e)	CosmicFlows-4 bulk-flow direction (297ř, −6ř)	CF4 (archival); DESI peculiar velocities	Cold-attractive: 161ř off (fails); hot-attractive: 19ř; sign mechanism unspecified

Detection of a particle dark-matter candidate, or an expansion history irreconcilable with the bounce-and-ejection mechanism, falsifies H1. Absence of the polarization ring at CMB-S4 sensitivity eliminates H2; absence of anisotropic sub-solar mergers and memory strain at ET/LISA sensitivity eliminates H3 and H6; a null in deeper spot statistics eliminates H4 and H7; H5 has no positive detection to lose. The polarization ring and single-event memory strain are out of reach until Simons Observatory / CMB-S4 and ET/LISA respectively, which set the 2030s decision timeline.

12. Limitations and Open Problems

1. **Hard-sphere ansatz.** Treating Planck relics as incompressible spheres is a modeling assumption motivated by the Compton–Schwarzschild coincidence, not a quantum-gravity result.
2. **Bounce mechanism and horizon structure.** The rebound requires the LQC-form modified Friedmann equation; the identification $\rho_c = \eta_{\rm rcp}\rho_P$ is a conjecture. A finite ball at this density is forty orders inside its Schwarzschild radius, so the bounce is meaningful only for a whole or super-horizon patch, and a trapped-surface timing analysis, together with the multi-bubble assembly timescale ($\gtrsim 10^{20}\,t_P = 5.4\times10^{-24}$ s), remains to be done.
3. **Spectral index.** ν_t and the relation $n_s - 1 = -\nu_t/3$ are fitted inputs; the compatibility of a Kolmogorov cascade with near scale invariance is unproven.

4. **Dark-matter fraction and temperature.** The turbulent kick distribution at $(\bar{f}, \sigma_{\rm rel}) = (0.7, 1.0)$ yields $88\% \pm 4\%$ ejection, consistent with the observed fraction, but the cascade model reaches $\sigma_{\rm rel} \approx 1$ only marginally and the end-to-end chain is untested. The ejecta are relativistic, hence hot rather than cold dark matter unless a redshifting or cooling argument reduces their free-streaming length by matter–radiation equality. Planck-relic dark matter must also face femtolensing, capture, and abundance constraints.
5. **Expansion history.** Neither the Newtonian toy nor the relativistic LTB lightcone produces apparent acceleration ($q_0^{\rm eff} = +0.16$ against ≈ -0.55), the toy's $w(z)$ drift runs opposite to the DES+DESI direction, and the kSZ gate caps any ejecta contribution at $\sim 1\%$ of dark energy. The adopted expansion history (Sec. 5.2) is therefore the pool model plus a constant tension of space; the framework does not predict the value of that tension and inherits the 10^{-123} fine-tuning of ΛCDM. It is also not yet reconciled with BBN abundances or the CMB acoustic peaks.
6. **Causal and relativistic structure.** The jammed patch is $\sim 10^{20}$ Planck lengths across, so homogeneity must be assumed (the horizon problem stands). The ejection phase proceeds where Newtonian escape velocity vastly exceeds c; the N-body mechanism lacks a relativistic formulation, which is prerequisite for H1 to be a physical theory rather than an analogy.
7. **Relic-to-plasma conversion.** The furnace must convert relics to Standard-Model plasma with the observed baryon asymmetry and photon-to-baryon ratio while ejected relics stay stable; no channel is specified.

Each item maps onto the observational program of Sec. 11; the theory is offered as a falsifiable mechanical alternative, not a completed replacement for ΛCDM.

Epilogue: Three Questions, Three Answers

We set out with three questions and a pool table. Here is where the balls came to rest.

How does it work? Picture the moment before the first moment: a false vacuum, tense as a drawn bow. Somewhere a bubble of true vacuum nucleates, its wall sweeping outward, catalysing more bubbles, until the whole patch is a jammed crush of 10^{60} Planck-mass spheres pressed to the random-close-packing limit. Nothing can fall further. A loop-quantum bounce fires, the pile rebounds, and the rebound is chaotic: $88\% \pm 4\%$ of the spheres are hurled outward, the fastest furthest, sorting themselves into shells like shot from a gun. That flying debris is our dark matter, and its velocity ordering is our Hubble flow. But the debris does not push the universe faster; nothing outside a volume ever can, and nothing inside it pushes either. The acceleration we measure comes from a constant tension of space, $w = -1$, which we adopt and cannot yet derive. So the model works as far as the matter goes, hands the expansion to a tension it does not explain, and carries three confessed wounds: the ejecta are born relativistic and must somehow cool, the thermal ring we predicted around the Cold Spot is not there, and the bounce is only meaningful if the jammed patch was the whole universe rather than a ball inside its own horizon.

Where are we? Inside that debris cloud, somewhere between the surviving core and the outermost shell. Only the outermost layers keep any memory of the initial geometry; everything deeper has been mixed. We looked for our own displacement from the centre in the galaxy velocity field, and after the mock-mask and 2M++ controls the answer is: no measurable offset, a few megaparsecs at most, direction undetermined. What we do have is a bulk flow of about 400 km s^{-1} that the kinetic Sunyaev–Zel'dovich data cap near 600, and a sky whose extreme spots all lie in the southern hemisphere. The honest reading is that we are not visibly off-centre, and that the lopsidedness of the sky is more plausibly the work of a neighbour than of our own position. Where we are remains the least answered of the three questions, and the paper says so.

Where is the second Big Bang? Follow the coldest patch of the oldest light: galactic longitude 203ř, latitude −56ř, the Cold Spot. If it is the fingerprint of a neighbouring Big Pool Bang pressing on our last-scattering surface, then that neighbour sits just beyond it, about one to two billion light-years past the edge of everything we can see, with a mass of order 10^{48} to 10^{50} kg. Seven other extreme spots line up on the same side of the sky, a candidate pool of neighbours clustered rather than surrounding us. The neighbour reading makes promises it can be held to. It must leave a curl-mode fingerprint at the spots, or not (P9). Its $1/r$ potential must leak a few microkelvin across the whole sky in a pattern fixed by the spots (P10); the first unmasked look found no correlation. Its pull must point along the bulk flow (P11); with cold spots attractive it points the wrong way, and the model is not allowed to flip the sign to suit the data. Each of these is cheap, archival, and decisive. The neighbour hypothesis will not survive as a story; it will survive, or fall, as a set of numbers.

That is the adventure in one paragraph: a jammed crystal that bounced, a cloud of debris we live inside, and a cold fingerprint on the far wall that may belong to someone else. The next moves are already on the table.

Appendix A. Exploratory Null Results and a Pre-registered Test

This appendix records the exploratory battery run on the eight most extreme large-scale spots of the Planck SMICA map ($N_{\text{side}} = 128$, $|b| > 25$ř, 2ř smoothing; per-spot amplitude ΔT and half-maximum radius θ; positions in Sec. 10). Nulls are 60–300 Gaussian realizations drawn from the map's own pseudo-C_ℓ and passed through the identical pipeline unless stated otherwise.

Locus test. Log–log fit of $|\Delta T|$ against θ: slope -0.00, $r = -0.01$, 27th percentile of the Gaussian null ($+0.07\pm0.10$). No amplitude–size correlation is detected, but the test is powerless as implemented (radii quantized to 2.5ř–5.5ř, amplitude spread $\sim$10%). It is reported as unresolved pending injection-recovery validation, not as an exclusion. Neighbours at a range of distances would populate a diffuse cloud rather than a locus, and one genuine neighbour among seven Gaussian impostors would not move the fit.

Internal dipoles and spin budget. Monopole-plus-dipole fits within 6ř in the local tangent frame: the eight extremes split 4/4 hot/cold with rank-matched amplitudes summing to $-11\,\mu$K against a mean $|\Delta T| \approx 162\,\mu$K (7% of Gaussian skies as balanced); opposite-sign spots lie 4.0ř closer than same-sign (null $+5.2$ř $\pm$ 16.3ř, $p = 0.30$); nearest-neighbour spacing is no more regular than Gaussian. Converting dipoles to spin-axis proxies ($\mathbf{L} \propto \hat{c} \times \mathbf{d}$), the Cold Spot's internal dipole ($16.4\,\mu$K/deg) is the largest by a factor 2.3 and the others weakly co-align. These fits are dominated by $\ell \sim 2$–20 gradients that Gaussian peaks generically carry and that are correlated across co-hemispheric spots, so they are upper bounds on rotation, not spin measurements; a gradient-subtracted re-fit is required before any spin fraction can be quoted.

Spiral winding. The azimuthal phase of the $m = 1$ mode in 1.5ř rings from 1.5ř to 15ř around the Cold Spot is locked ring-to-ring with coherence 0.988 — above all 300 random positions (null 0.72 ± 0.18) — and drifts by $\sim$ 3ř of phase per degree of radius ($\sim$ 40ř total twist). A static background gradient gives high coherence but zero winding; only the winding is spin-specific. The $m = 2, 3$ modes are unremarkable, and this is a post-hoc test.

Curl search. The public PR4 lensing release contains only gradient modes, so we implemented an approximate temperature-only quadratic estimator ($N_{\rm side} = 256$, $\ell \leq 512$, spin-1 curl projection, mean field from 40 Gaussian simulations; unnormalized, valid only as a relative comparison). Stacking curl-potential variance in 6ř discs: the eight spots sit at the 82nd percentile of 60 simulated skies and the Cold Spot at the **38th** — indistinguishable from a random position. This does not trigger the retirement clause of P9, which is bound to a sensitivity reaching the temperature-dipole-implied rotation that this reduced-resolution estimator does not attain (Planck's full-depth reconstruction uses $\ell \leq 2048$ with polarization).

Polarization first look. On the SMICA Q/U maps ($N_{\rm side} = 256$, $\ell \leq 512$, E/B decomposition, 1ř smoothing): stacked B-mode variance in 6ř discs at the eight spots is at the 66th percentile of random positions, E-mode variance at the 35th, and the Cold Spot's radial-E profile is consistent with zero in every 2ř ring to 20ř with per-ring errors of 0.6–2.7 μK — the predicted 1–3 μK ring (P1) is neither detected nor excluded.

Lensing convergence. The sign-weighted PR4 κ stack on the eight positions is $+0.006$ against a random-position null of 0.000 ± 0.013 (4/8 sign matches): no integrated mass scars. The Cold Spot's inner-2ř $\kappa = -0.070$ vs null $+0.013 \pm 0.042$ (-2σ) has the sign of its known shallow foreground supervoid; the 8ř–20ř annulus never exceeds $+0.6\sigma$, so there is no mass "belt". A stack on $\sim$27,000 ordinary small spots (1ř smoothing, $|T| > 1.5\sigma$) shows cold spots on κ deficits (-7.2σ) and hot on excesses ($+2.2\sigma$), but κ is reconstructed from the same temperature data and quadratic estimators leak temperature extrema into the mass map; whether this is leakage or genuine ISW$\times$lensing is undiscriminated, and a polarization-only κ stack is the decisive check we have not performed.

CMB-independent tracers. Against the Quaia Gaia–unWISE quasar catalogue (756k quasars, $z \sim 1$–3, selection-function corrected) the small-spot correlation vanishes (cold -1.1σ, hot -0.3σ) and the eight extremes are within $\sim 1\sigma$ of random (Cold Spot -0.018, 0.6σ); this is not a sensitive null, since Quaia's kernel barely overlaps the ISW kernel. Against WISE$\times$SuperCOSMOS (17.7M galaxies, $z = 0.1$–0.4, high-passed at 10ř): small spots remain null (cold -0.7σ, hot $+0.2\sigma$), while the extremes show seven of eight sign matches and a sign-weighted stack of $\approx +2.7\sigma$ against a correlated-noise null from 300 random 5ř discs. Qualifications: the result rests on three sightlines ($|z| \approx 2.0$–2.3); the nominal binomial 3.5% ignores the three-tracer, multi-statistic search (global $p \sim 10$–30%); extremeness-conditioning inflates the recovered correlation (Eddington-type bias); the 10ř high-pass self-subtracts genuine $\gtrsim$ 10ř supervoids; and extinction/stellar residuals at $|b| = 25$–40ř are of order the per-spot signals. Verdict: a 1–2σ-class hint that the extremes are partly sourced by $z \sim 0.1$–0.4 superstructures, to be confirmed or retired by a pre-registered re-run at two baseline scales with $|b| > 30$ř and extinction regression. The 2M++ cross-match of the eight sightlines ($< 180\,h^{-1}$ Mpc) matches spot signs at coin-flip rates (5 of 8).

JWST handedness. The JADES field at (224ř, -54ř), 8.6ř from the Cold Spot, shows a claimed 2 : 1 handedness asymmetry in 263 high-redshift galaxies (Shamir 2025). The sample is small, the method contested, and a Doppler boost from the Milky Way's rotation is a mundane candidate. The discriminating measurement is a control field — the identical pipeline on COSMOS or GOODS-N ($>$ 100ř away): a systematic reproduces the asymmetry everywhere; a Cold Spot spin column produces it only on the column. The local anchor is already measured: the same-handedness angular correlation over 77{,}840 SDSS galaxies (11.2M pairs, 0.05ř–4ř) is zero in every bin, with a 1.9σ global excess.

Statistical accounting. Across the battery (locus slope, hemisphere count, hot/cold balance, pairing distance, lattice regularity, dipole magnitudes, cancellation ratio, winding coherence, κ stack, curl percentile, tracer stacks) the individual results are $p \sim 0.06$–0.7; under any global look-elsewhere correction the sub-2σ items are jointly consistent with noise, and we say so plainly. Further method limits: the Gaussian nulls use the unmasked pseudo-C_ℓ without mask deconvolution, anisotropic noise, or foreground residuals (end-to-end NPIPE simulations are the required upgrade for any sub-percent claim), and the spots were selected on the same map on which all statistics were evaluated, so the hypotheses were partly formulated post hoc. What survives is qualitative: the extreme-spot sky is one-sided at literature-level significance, the Cold Spot is the outlier in every channel examined, and nothing requires a population of additional objects.

Pre-registered commitment (P9). The load-bearing, falsifiable claim of the vortex reading is the curl test. We commit in advance that a clean stacked curl-mode null at the Cold Spot position, at a sensitivity reaching the temperature-dipole-implied rotation, *retires* the vortex interpretation of that direction in favor of the non-rotating neighbour reading (H3/H6) — not merely disfavors it. If curl variance is instead detected localized there, the vortex reading stands and the Cold Spot becomes a relic of our own bang's border shell rather than a neighbour's imprint.

Appendix B. A Fixed Grid of Surrounding Bangs

The Pool Model in its most symmetric form places Big Pool Bangs on a fixed lattice with our own at one site. We ask whether the neighbours' pull on our ejecta can contribute to the observed late-time acceleration. Point masses of equal mass were placed on a cubic grid of unit spacing (26 neighbours for $3 \times 3 \times 3$, 124 for $5 \times 5 \times 5$); test galaxies were placed at distance r from us in random directions and the vector sum of the neighbours' inverse-square pulls was evaluated. Units: the pull of one neighbour on us.

r (grid units)	Sky-averaged outward pull	Toward a neighbour (axis)	Toward a gap (diagonal)
0.1	0.0000	+0.012	−0.008
0.2	0.000	+0.10	−0.06
0.3	0.000	+0.34	−0.20
0.4	0.00	+0.85	−0.44
0.45	0.00	+1.26	−0.60

The results are identical for both grid sizes. Galaxies lying toward a neighbour are pulled outward with a pull that grows steeply with r; galaxies lying toward the gaps between neighbours are pulled inward; the sky average is zero at every radius. This is Gauss's law, not a property of the arrangement: the sky-averaged radial acceleration at radius r is $GM_{\mathrm{enc}}(r)/r^2$, and external sources enclose nothing. Only mass, or negative pressure, *inside* the observed volume can change the isotropic expansion rate. The lattice therefore predicts (i) zero contribution to the monopole acceleration, and (ii) a direction-dependent expansion rate with the cubic-harmonic ($\ell = 4$) pattern of the lattice, fast toward the six nearest neighbours and slow toward the eight diagonal gaps. Observation (ii) is a further falsifier for H7: supernova and BAO expansion-rate anisotropy is limited to the few-percent level with no cubic pattern, while the amplitude implied by the neighbour masses and distances of Sec. 10e is $\sim 10^{-7}$ at the $r \approx 0.3$ reached by the supernova sample. The one form in which surrounding mass does raise the local expansion rate isotropically is a local underdensity, where the effect comes from the mass *missing inside*: the observed Keenan–Barger–Cowie underdensity ($\delta \approx -0.3$ within $\sim$300 Mpc) raises the local H_0 by a few percent, a candidate contribution to the

Hubble tension, while a Gpc-scale void deep enough to mimic Λ is excluded by the kSZ gate of Sec. 5. These conclusions are the basis for the adopted expansion history of Sec. 5.2.